Databases

IMPORTANCE OF DATABASES IN INFORMATION SYSTEMS, HISTORY OF DATABASES, TYPES OF DATABASES

Database

- A database is an organized collection of data or information.
- It is an organized collection, because in a database, all data is described and associated with other data.



Database contd/

- All information in a database should be related as well; separate databases should be created to manage unrelated information. For example, a database that contains information about students should not also hold information about company stock prices.
- Databases are not always digital – a filing cabinet, for instance, might be considered a form of database. For the purposes of this module, we will only consider digital databases.

Database contd/

- Data can be stored and accessed electronically or manually.
- A computer-based or digital database is accessed electronically while a manual database is accessed manually.
- A database allows for efficient storage, retrieval, and management of information.

Database contd/

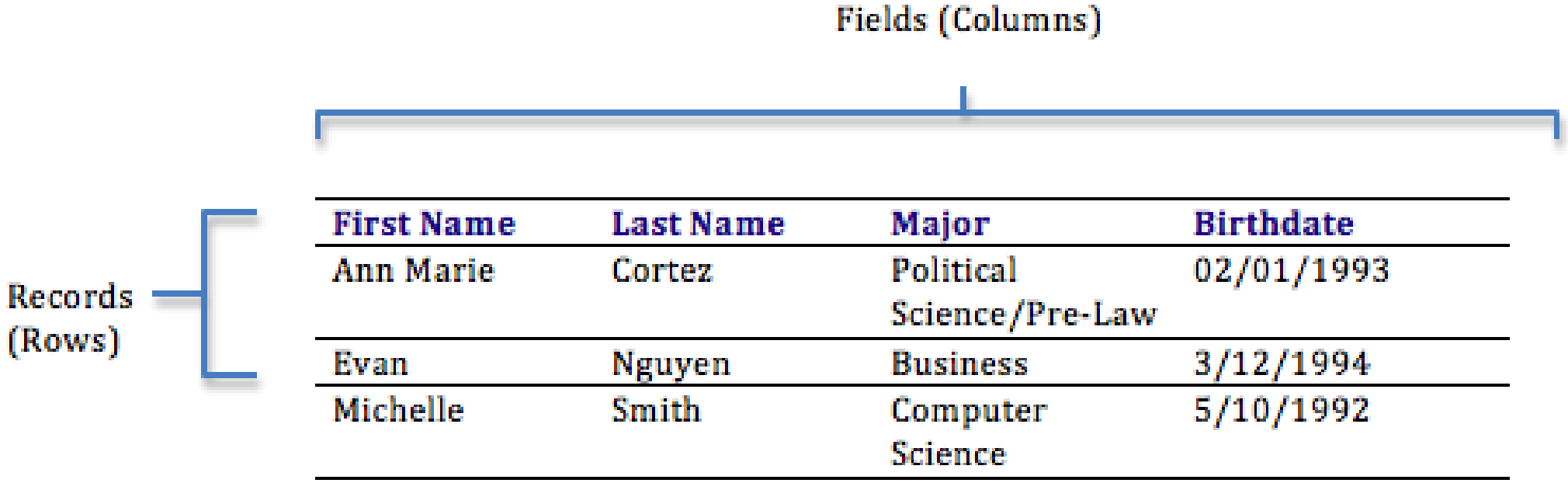
- The goal of many information systems is to transform data into information in order to generate knowledge that can be used for decision making.
- In order to do this, the system must be able to take data, put the data into context, and provide tools for aggregation and analysis. A database is designed for just such a purpose.

Database contd/

- Databases can be organised in many different ways, and thus take many forms.
- The most popular form of database today is the relational database. Popular examples of relational databases are Microsoft Access, MySQL, and Oracle.
- A relational database is one in which data is organized into one or more tables.
- Each table has a set of fields, which define the nature of the data stored in the table.
- A record is one instance of a set of fields in a table.

Database contd/

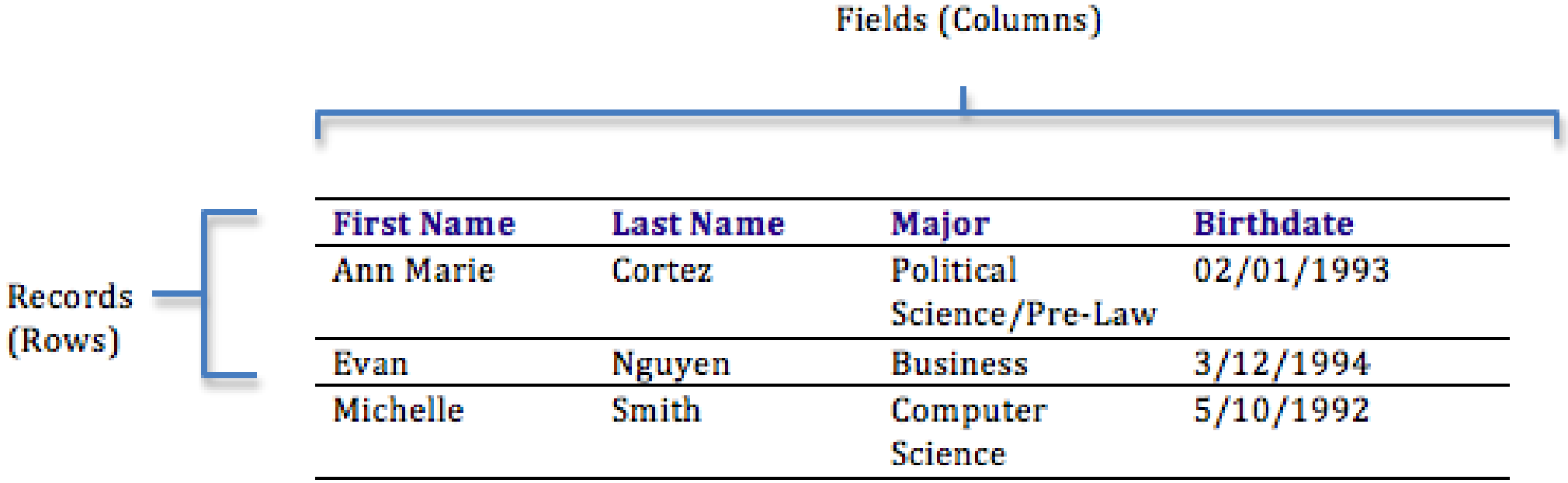
- In the example below, we have a table of student information, with each row representing a student and each column representing one piece of information about the student.



First Name	Last Name	Major	Birthdate
Ann Marie	Cortez	Political Science/Pre-Law	02/01/1993
Evan	Nguyen	Business	3/12/1994
Michelle	Smith	Computer Science	5/10/1992

Database contd/

- In the example below, we have a table of student information, with each row representing a student and each column representing one piece of information about the student.



First Name	Last Name	Major	Birthdate
Ann Marie	Cortez	Political Science/Pre-Law	02/01/1993
Evan	Nguyen	Business	3/12/1994
Michelle	Smith	Computer Science	5/10/1992

Importance of database

1. Improved Metadata Management
   - Metadata is all the information, such as the table's name, column titles, and data types. Good metadata management makes data easier to locate and utilize. A sound database leads to improved metadata management.
   - Users may quickly discover the required data when everything is organized correctly without browsing multiple files. A database can also maintain metadata and keep track of changes being made in the data. This is significantly important when data is often altered while metadata is always up-to-date.

Importance of database contd/

2. Improved Data Security
   - A database helps improve data security by providing various features making the database difficult for unauthorized access.
   - The database can be used to
     i. ensure that only authorized users access data
     ii. prevent unauthorized individuals from accessing data.
     iii. allow administrators to know which users have access to which data.
   - This information helps in identifying potential security breaches and taking corrective action.

Importance of database contd/

3. Reduce Data Redundancy
   - A database eliminates data redundancy and inconsistency by reducing the number of isolated files stored in the system as duplicate data.
   - It does not allow business authorities to eliminate data redundancy but provides controllability to govern data redundancy

Importance of database contd/

4. Increased end-user productivity
   - The availability of data, combined with the tools that transform data into usable information, empowers end users to make quick, informed decisions that can make the difference between success and failure in the global economy.
5. Data Integrity
   - Data integrity means data is consistent and accurate in the database. It is essential as there are multiple databases in DBMS. All these databases contain data which is visible to multiple users. Therefore, it is essential to ensure that data is consistent and correct in all databases for all users.

Importance of database contd/

5. Better data integration
   - Wider access to well-managed data promotes an integrated view of the organization’s operations and a clearer view of the big picture. It becomes much easier to see how actions in one segment of the company affect other segments.

History of Databases

- The history of databases dates back long before computers were invented.
- In the past, data was stored in journals, libraries, and filing cabinets, taking up space and making it difficult to find and back up.
- The advent of computers in the early 1960s marked the beginning of computerised databases.
- Charles Bachman designed the first database known as the Integrated Data Store, followed by the Information Management System developed by IBM.

History of Databases contd/

- These databases were navigational, requiring users to navigate through the entire database to find the information they wanted.
- The two main models of this navigational database were the hierarchical model developed by IBM and the network model introduced at the Conference on Data Systems Languages (CODASYL)

History of Databases contd/

- In 1970 E. F. Codd wrote a paper titled "A Relational Model of Data for Large Shared Data Banks." This paper marked the beginning of the relational database, which shows the relationship between different data records and is more space-efficient, making it cost-effective for data storage.
- His ideas changed the way people thought about databases. In his model, the database's schema, or logical organization, is disconnected from physical information storage, and this became the standard principle for database systems.

.

History of Databases contd/

- It led to the creation of INGRES, a relational database model, at the University of California, Berkeley, which used the QUEL query language.
- IBM released its own relational database, System R, which was the first to use Structured Query Language (SQL).
- In the 1980s marked a time of growth and standardisation for relational databases, with the navigational database models fading.
- The commercialisation of relational systems saw a rise in their use and popularity, with SQL becoming the standard language for databases.

History of Databases contd/

- Another noteworthy event was the emergence of Object-oriented database management systems (OODBMS), which viewed data as objects and worked with programming languages that supported the object-oriented approach.
- The 1990s saw the rise of the World Wide Web, fueling demand for client-server database systems, and the exponential growth of the database industry.
- MySQL was created in 1995, offering an alternative to the database systems of large companies like Oracle and Microsoft. Object-oriented database systems also grew more popular in the 1990s.

History of Databases contd/

- The term NoSQL was coined in 1998, referring to databases that use a query language other than SQL to store and retrieve data.
- NoSQL databases are useful for unstructured data and saw growth in the 2000s, allowing for faster processing of larger, more varied datasets.
- NoSQL databases are more flexible than the traditional relational databases that had risen the decade before.
- The 2010s saw a rise in data awareness, with the emergence of big data and increased emphasis on data protection. was highlighted by legislation like GDPR and the NIS directive.

History of Databases contd/

- In conclusion, the history of databases is a rich and evolving one that has impacted and been impacted by the advancements of technology, and the need to manage and protect large amounts of data efficiently and effectively.

Types of Databases contd/

- **There are several types of databases, including:**

1. **Relational databases (RDBMS)**
   - ✓ Organise data into tables with well-defined relationships. Examples include MySQL, Oracle.Relational databases are the other major type of database, opposite of NoSQL.
   - ✓ A good representation of a relational database would be the connection between a person shopping online and their shopping cart. Relational databases are often preferred when you are concerned about the integrity of your data, or when you are not particularly focused on scalability.

Types of Databases

2. **NoSQL databases**

- ✓ Store data in a variety of formats, such as key-value, document, or graph. Examples: MongoDB, Cassandra.
- ✓ A NoSQL database has a hierarchy similar to a file folder system and the data within it is unstructured, or non-relational. This lack of structure allows them to process larger amounts of data at speed and makes it easier to expand in the future. Cloud computing regularly makes use of NoSQL databases.

Types of Databases

3. Object-oriented databases (OODBMS)

- ✓ Store data in the form of objects, similar to object-oriented programming. Examples: Gemstone, Matisse.
- ✓ Data is represented as objects and classes.
- ✓ An object is an item, such as a name or phone number, while a class is a group of objects. Object-oriented databases are a type of relational database.
- ✓ Consider using an object-oriented database when you have a large amount of complex data that you want to process quickly.

Types of Databases

4. **Graph databases**

- ✓ Designed to store and query complex relationships between data. Examples: Neo4j, Amazon Neptune.
- ✓ Graph databases are databases that focus equally on the data and the connections between them.
- ✓ Data is not constricted to predefined models.

Types of Databases contd/

5. **Time-series databases**
   - ✓ Optimized for storing and retrieving large amounts of time-stamped data or time series. Examples: InfluxDB, OpenTSDB.
   - ✓ Time series data are simply measurements or events that are tracked, monitored, downsampled, and aggregated over time. This could be server metrics, application performance monitoring, network data, sensor data, events, clicks, trades in a market, and many other types of analytics data.

Types of Databases contd/

5. **Centralized database**
   - ✓ A centralized database is one that operates entirely within a single location.
   - ✓ Centralized databases are typically used by bigger organizations, such as a business or university.
   - ✓ The database itself is located on a central computer or database system.
   - ✓ Users can access the database through a computer network, but it is the central computer that runs and maintains the database.

Types of Databases contd/

6. **Cloud database**
   - ✓ A cloud database is one that runs over the Internet.
   - ✓ The data is stored on a local hard drive or server, but the information is available online.
   - ✓ This makes it easy to access your files from anywhere, as long as you have an Internet connection.
   - ✓ To use a cloud database, users can either build one themselves or pay for a service to store their data for them.
   - ✓ Encryption is an essential part of any cloud database, as all information needs to be protected as it is transmitted online.

Types of Databases contd/

7. **Distributed databases**
   - ✓ Span multiple physical locations, providing high availability and fault tolerance. Examples: Google Spanner, CockroachDB.